Runbook: Rolling Upgrade of Ledgerline Gateway 4.1 to 4.2

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Revision	3 (19 October 2026)
Owner	Platform Operations
Review cycle	Every six months
Applies to	Production clusters eu-west-a and eu-west-b

1. Purpose and scope

This runbook describes how to upgrade the Ledgerline payment gateway from version 4.1.x to 4.2.0 without interrupting card authorisations. The procedure is a rolling upgrade: nodes are drained, upgraded and returned to the pool one at a time. It does not cover the database schema migration, which is handled by runbook OPS-RB-109.

2. Prerequisites

- A change ticket in the Approved state, with the maintenance window recorded.
- Shell access to the bastion host with the ops-deploy role.
- The release bundle ledgerline-4.2.0.tar.gz, verified against its published checksum.
- At least four healthy nodes in the target cluster before you begin.

3. Pre-flight checks

Run every check below from the bastion host. Stop and escalate if any result differs from the expected value.

Check	Command	Expected result
Cluster health	`llctl cluster status`	All nodes READY
Queue depth	`llctl queue depth --all`	Below 500 messages
Disk space	`df -h /opt/ledgerline`	At least 5 GB free
Certificate expiry	`llctl tls expiry`	More than 30 days

4. Procedure

Repeat steps 4.1 to 4.3 for each node in turn, starting with the node that currently carries the least traffic.

4.1 Drain the node

Remove the node from the load balancer and wait until in-flight transactions have completed.

```
llctl node drain gw-node-03 --timeout 300
llctl node status gw-node-03
# wait for: state=DRAINED active_sessions=0
```

4.2 Install the new release

```
sudo systemctl stop ledgerline-gateway
sudo tar -xzf /srv/releases/ledgerline-4.2.0.tar.gz -C /opt/ledgerline
sudo ln -sfn /opt/ledgerline/4.2.0 /opt/ledgerline/current
```

Version 4.2 introduces a new connection pool section in the gateway configuration. Update /etc/ledgerline/gateway.yaml so that it contains the following block before restarting the service.

```
gateway:
  listen: 0.0.0.0:8443
  pool:
    max_connections: 256
    idle_timeout_s: 90
```

```
    health_check:
      path: /healthz
      interval_s: 10
  tls:
    min_version: "1.2"
```

4.3 Return the node to service

1. Start the service with sudo systemctl start ledgerline-gateway.
2. Confirm that llctl node status reports version 4.2.0 and state READY.
3. Undrain the node with llctl node undrain gw-node-03 and check the dashboard:
   - authorisation error rate below 0.5 %
   - p95 latency below 300 ms
4. Wait ten minutes before moving on to the next node.

5. Rollback

If the error rate for an upgraded node exceeds 0.5 % for more than five minutes, roll that node back immediately. Do not continue the upgrade until the cause has been understood.

```
sudo systemctl stop ledgerline-gateway
sudo ln -sfn /opt/ledgerline/4.1.7 /opt/ledgerline/current
sudo systemctl start ledgerline-gateway
```

6. Contacts

Role	Name	Extension
Duty platform engineer	On-call rota (see PagerHub)	4410
Payments product owner	Selin Arkwright	4127
Database on-call	On-call rota (see PagerHub)	4415